



COMPONENTS

Ultra-miniature reed switches

Littelfuse has introduced the MITI-7 series of ultra-miniature reed switch, a Form A (single-pole, single-throw normally-open) switch that is hermetically sealed in a 7mm-long glass envelope.

These offer sensitivity options ranging from 6AT to 20AT. Typical applications include position and speed sensing in a broad array of markets such as appliances, building and home automation, data centres, industrial, and healthcare.

Littelfuse
www.littelfuse.com

Quadband VCOs

This new quadband voltage controlled oscillator (VCO) series offers wideband capabilities in today's RF and microwave environments without compromising phase noise performance.

These deliver a wide RF response, high frequency flexibility (8.3GHz to 26.6 GHz), low phase noise, and consume low current. The VCOs do not generate subharmonic frequency tones, making them suitable for high-end instrumentation, and aerospace and defence applications.

Analog Devices
www.analog.com

Ultra-low-power microcontroller

The RA2L1 microcontroller from Renesas provides ultra-low power and an innovative touch interface that makes it ideal for home appliances, industrial and building automation, medical and

healthcare, and consumer human-machine-interface (HMI) IoT applications. The general-purpose RA2L uses the Arm Cortex-M23 core, operating up to 48MHz, and consumes low power. It includes capacitive touch sensing, embedded flash memory densities up to



256kB, SRAM at 32kB for analogue communications, timing peripherals, and safety and security functions.

Renesas Electronics Corporation
www.renesas.com

Space-qualified FPGA

RT PolarFire Field Programmable Gate Array (FPGA) from Microchip enables high bandwidth for on-orbit processing systems, low power consumption, and the ability to withstand radiation effects in space. The RT PolarFire FPGA increases the computational performance of satellite payloads and makes optimal use of limited downlink bandwidth. It can also withstand Total Ionising Dose (TID) exposure beyond the 100 kilorads (kRads) typical of most earth-orbiting satellites and many deep-space missions.

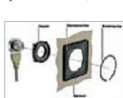
Microchip Technology
www.microchip.com



Connectivity solution

The new connectors from Fischer allow design engineers to further optimise cable management in line with their SWaP (size, weight, and power) requirements. These integrate low-profile connectors, cable assemblies, and active devices easily into all sorts of materi-

als—even the most flexible of fabrics for heavy-duty tarp cover/tent, sail, vehicle tyre blankets, and smart backpacks.



Their adaptor and retaining ring facilitate quick fit and interchange of receptacles. Applications include defence, medical, industrial, wearables, and IoT.

Fischer Connectors
www.fischerconnectors.com

Fuel-gauge IC

The new MAX17320 high-accuracy fuel-gauge and protection IC keeps in check self-discharge hazards for up to 2 or 4 series Lithium-ion (Li+) cells (2S-4S). The device is part of an IC family equipped with Model-Gauge m5 EZ algorithm that delivers accurate state-of-charge readings, eliminating the need for battery characterisation for most common Li+ cells. This fuel gauge also offers low quiescent current and features SHA-256 authentication for safeguarding systems from counterfeit batteries. The MAX17320 is suitable for hand-held devices with 2S-4S Li+ batteries for such applications as computing, IoT, power tools, consumer, healthcare, and mobile devices, which can be met with high customer safety standards.

Maxim Integrated
www.maximintegrated.com



Wireless-charging IC

A fast Qi wireless charging IC, the STWLC88, delivers 50W of power that enables smartphones, tablets, laptops, and other personal electronic devices to charge without cables in a safe, quick, and

